SETUP ANALYSIS RESULTS

DOES THE WEATHER AFFECT MY BF`S HEADACHES?





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QUESTION:

My BF gets recurring headaches
and wonders whether
the weather might play a role?

Sounds great. But data first!



#Data Source

So I combined his headache diary from **Migrebot**
with weather data from **Open-Meteo.**

Let's figure it out!



image generated by AI

THE DATA

After merging the two datasets,
I ended up with 176 days:

38
with headaches

138
without

**That's a headache on 21.6% of days —
more than one in five!**

As the person sharing a home with him,
I can confirm: that feels like a lot.

BORING STUFF

For each weather variable, I tested whether it differed between headache and non-headache days.

H_0: No difference in the weather characteristic being tested between headache and non-headache days.

H_1: A difference exists.

SIGNIFICANCE LEVEL: $\alpha = 0.05$

I mainly used **Welch's t-tests**, with separate tests for wind direction and weather categories.

I then applied **Holm correction** to account for multiple comparisons.

For all but one of the comparisons, the adjusted p-value was above 0.05, so **I did not reject H_0.**

	weather_metric	test	p_value_raw	p_value_holm	reject_h0_holm
15	wind_direction_10m_dominant (°)	Wheeler–Watson	0.000001	0.000022	True
9	dew_point_2m_mean (°C)	Welch t-test	0.004547	0.072756	False
14	wet_bulb_temperature_2m_mean (°C)	Welch t-test	0.006833	0.102489	False
1	apparent_temperature_mean (°C)	Welch t-test	0.010019	0.140262	False
0	temperature_2m_mean (°C)	Welch t-test	0.019680	0.255837	False
10	relative_humidity_2m_mean (%)	Welch t-test	0.061970	0.743636	False
13	wind_speed_10m_mean (km/h)	Welch t-test	0.104395	1.000000	False
12	wind_gusts_10m_mean (km/h)	Welch t-test	0.126854	1.000000	False
11	pressure_msl_mean (hPa)	Welch t-test	0.281329	1.000000	False
8	cloud_cover_mean (%)	Welch t-test	0.840769	1.000000	False
6	precipitation_sum (mm)	Welch t-test	0.832582	1.000000	False
5	sunshine_duration (s)	Welch t-test	0.253049	1.000000	False
4	daylight_duration (s)	Welch t-test	0.504453	1.000000	False
3	et0_fao_evapotranspiration (mm)	Welch t-test	0.847133	1.000000	False
2	shortwave_radiation_sum (MJ/m²)	Welch t-test	0.923516	1.000000	False
7	precipitation_hours (h)	Welch t-test	0.589258	1.000000	False
16	weather_group	Chi-square	0.506227	1.000000	False

ONE SUSPECT LEFT: WIND DIRECTION

Wind direction was the only variable whose initial result remained statistically significant after Holm correction.

However, **a follow-up logistic model with a simple time trend and HAC standard errors did not provide sufficient evidence of an association (p = 0.205).**

Different methods, different assumptions – so no firm verdict on wind

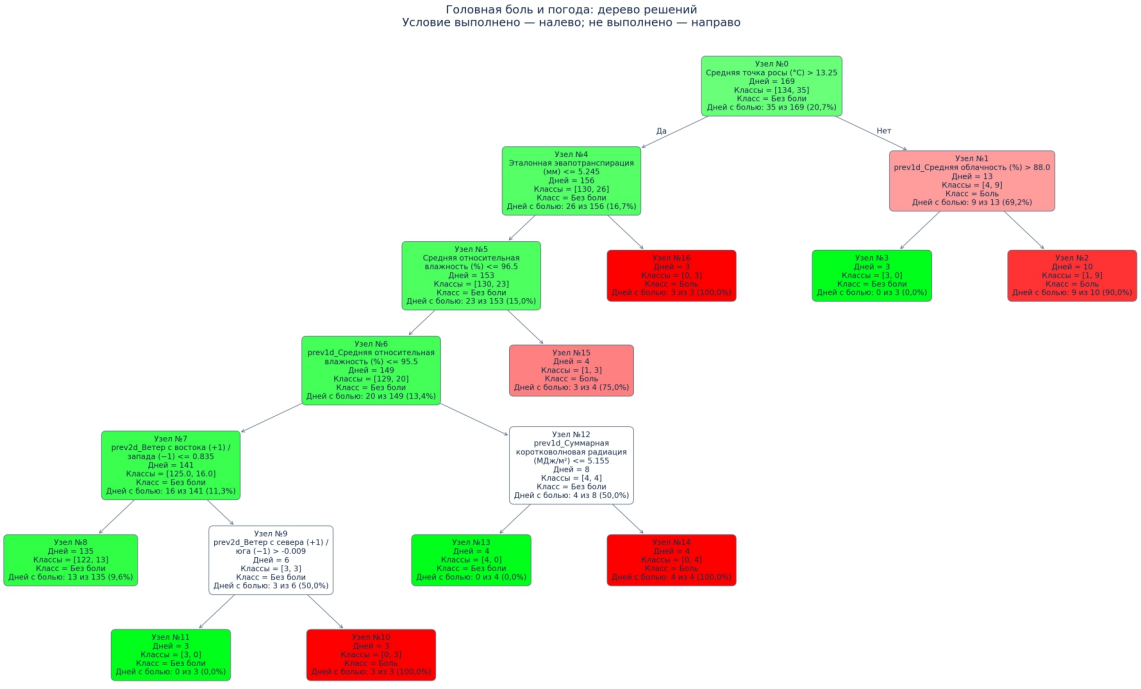
Wind direction	No headache (138 days)	Headache (38 days)
North	2.9%	2.6%
Northeast	1.4%	0.0%
East	2.2%	2.6%
Southeast	**5.1%**	**13.2%**
South	**7.2%**	**18.4%**
Southwest	26.1%	26.3%
West	**39.9%**	**23.7%**
Northwest	15.2%	13.2%

	hac_lag	n_days	wald_statistic	df	p_value_raw
0	3	176	3.737852	2	0.154289
1	7	176	3.172028	2	0.204740
2	14	176	3.283907	2	0.193602

CALLING IN THE ML DEPARTMENT!

Also known as my bf

My BF works in ML, so he also tried a decision-tree model



The tree found several combinations of weather conditions.
But some branches contained only three or four days.
“100% headache days” sounds impressive until you realise it means 3 out of 3.

The diagram alone cannot tell us whether these patterns would work on new days.

SO, IS THE WEATHER INNOCENT?

Not proven **guilty**

Not proven **innocent either**

This dataset did not give us enough consistent evidence to draw a firm conclusion about the relationship between weather and his headaches.

That is not the same as showing that weather has no effect.

ARE WE DONE?

Not quite!

From living with him, I've noticed that **strong smells and physical exertion** sometimes seem to go along with his headaches.

> *A long bike ride in the Turkish heat, for example. Romantic? Maybe.*
> *A useful diary entry?*
> ***Definitely!***

#These are personal observations, not confirmed triggers.

So the next step is a more detailed diary: when the pain starts, what he was doing beforehand, any strong smells, pain intensity and duration, and accompanying symptoms.

Blame the weather for your headache?
Sounds great. But data first.